Deploy

WORKSPACE

- Executive Overview
- Geospatial Intelligence
- Historical Analytics
- AI Model Insights
- Prediction Centre
- AI Farming Assistant

PILOT REGION

Punjab & Haryana

SATELLITE SOURCE

NASA FIRMS · VIIRS

DEVELOPED BY

Team TriVortex

PLATFORM STATUS

● **Operational**

Sustainability Hackathon · IIT Indore

EXECUTIVE DASHBOARD

Agricultural Fire Intelligence

Satellite observations and machine learning for historical fire analysis across Punjab and Haryana.

Platform Performance

SATELLITE DETECTIONS	AVERAGE FIRE POWER	OBSERVATION DAYS	TOP 10% FRP RECORDS
416,358	**6.38 MW**	**1,442**	**41,676**
NASA VIIRS observations	Mean FRP	Unique observation dates	Highest FRP observations

Intelligence Capabilities

Satellite Monitoring
Analyse real NASA satellite fire observations.

Hotspot Intelligence
Identify geographical areas with repeated detections.

Predictive Analytics
Estimate next-day satellite detection likelihood.

Historical Activity

Monthly Fire Detection Distribution



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416,358
NASA VIIRS observations

6.38 MW
Mean FRP

1,442
Unique observation dates

41,676
Highest FRP observations

Intelligence Capabilities

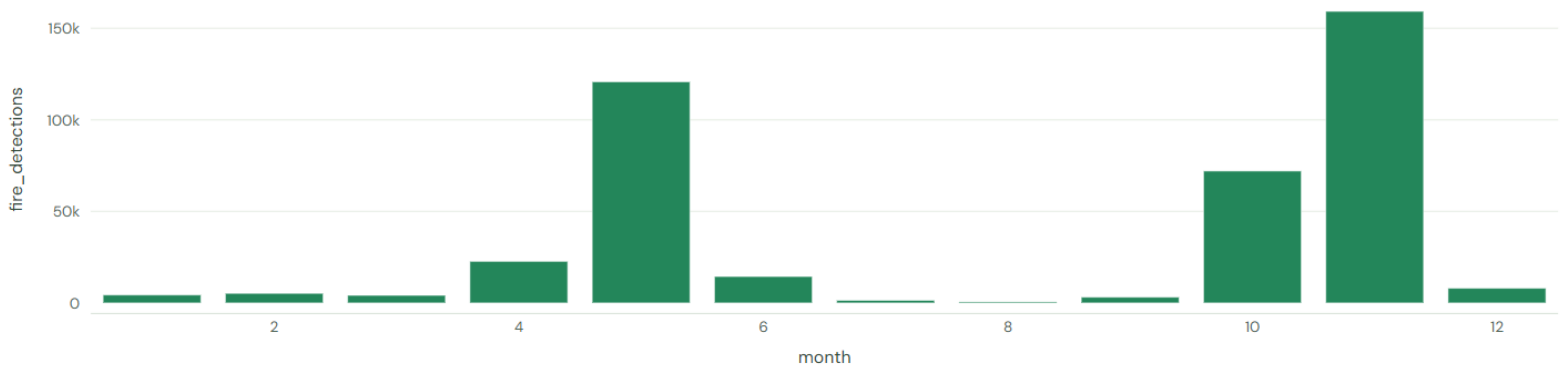
Satellite Monitoring
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NASA satellite detections are not automatically confirmed agricultural residue-burning incidents.

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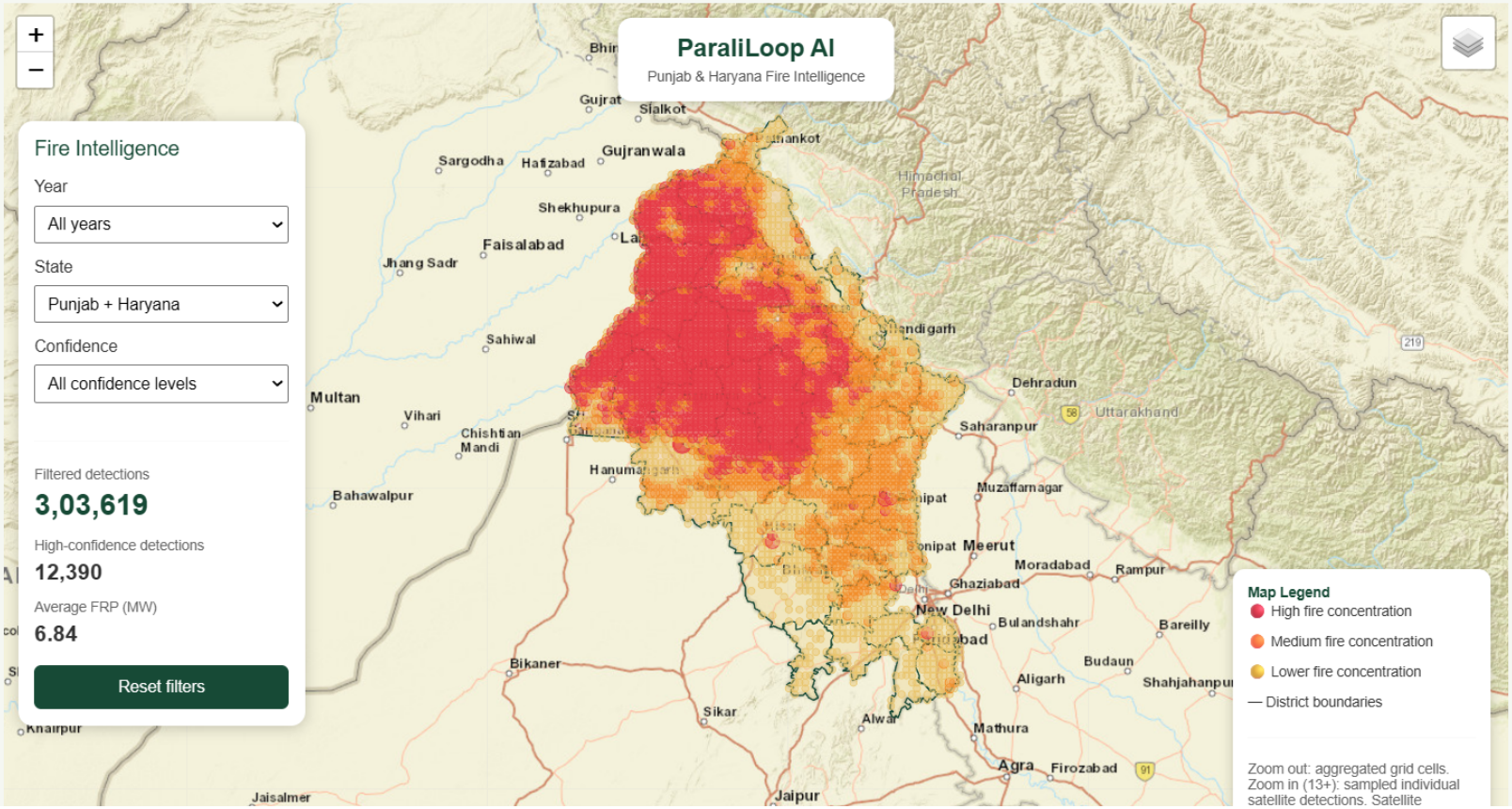
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LOCATION ANALYTICS

Geospatial Intelligence

Explore geographical concentrations of historical satellite fire detections.



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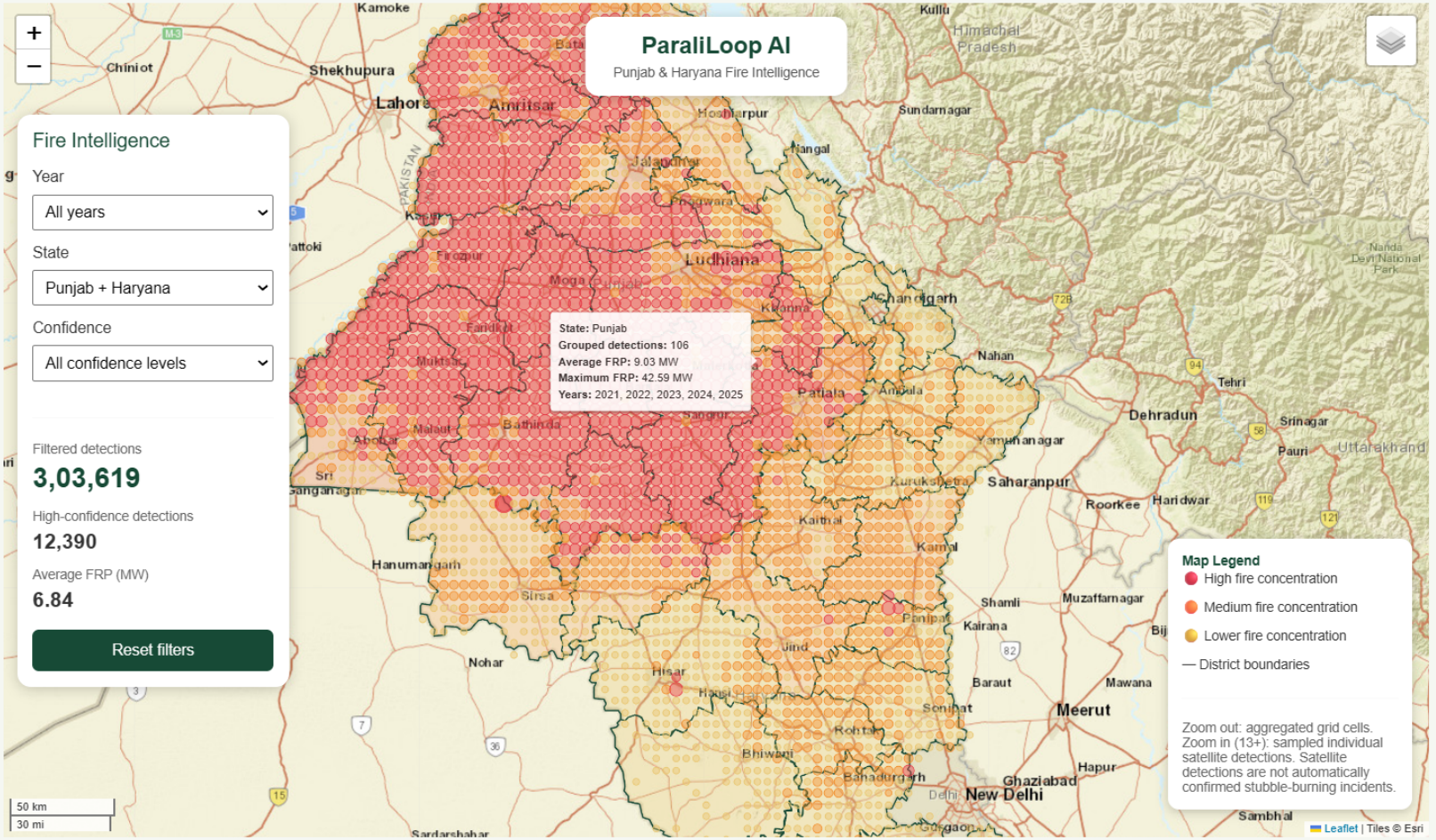
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Map represents historical satellite detections. It does not independently confirm stubble burning.

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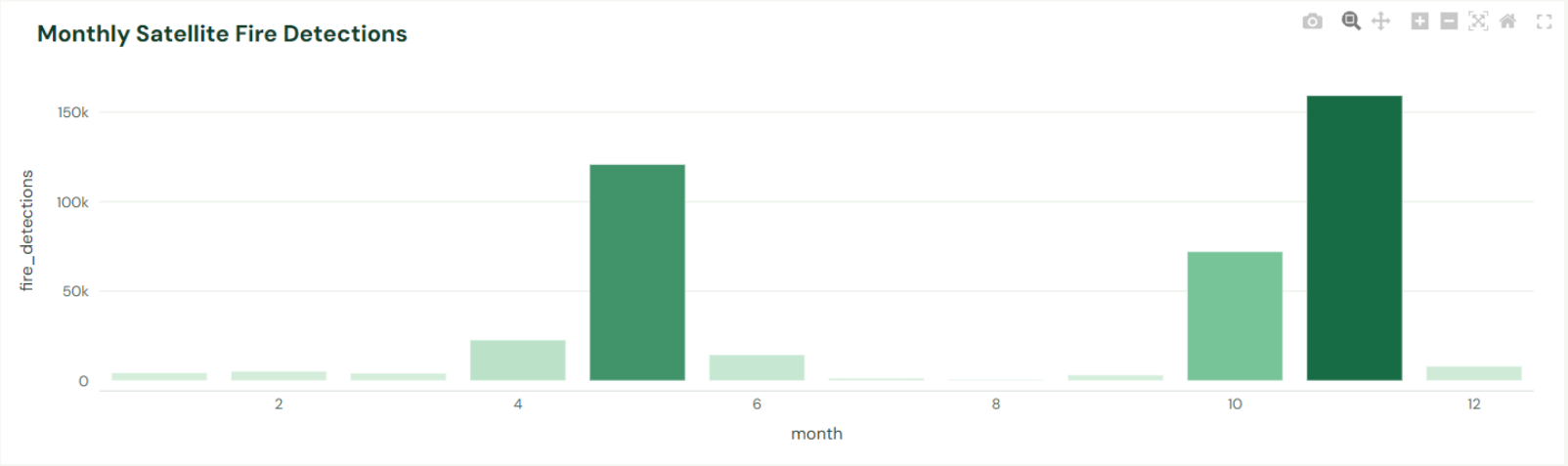
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DATA EXPLORATION

Historical Analytics

Discover seasonal patterns, annual variation and fire intensity.

Monthly Detection Distribution



Annual Detection Distribution



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MACHINE LEARNING

AI Model Insights

Evaluate the performance of our machine learning prediction model.

Model Evaluation

ACCURACY	PRECISION	RECALL
82.57%	**32.27%**	**54.48%**
Overall correct predictions	Positive prediction reliability	Positive cases identified

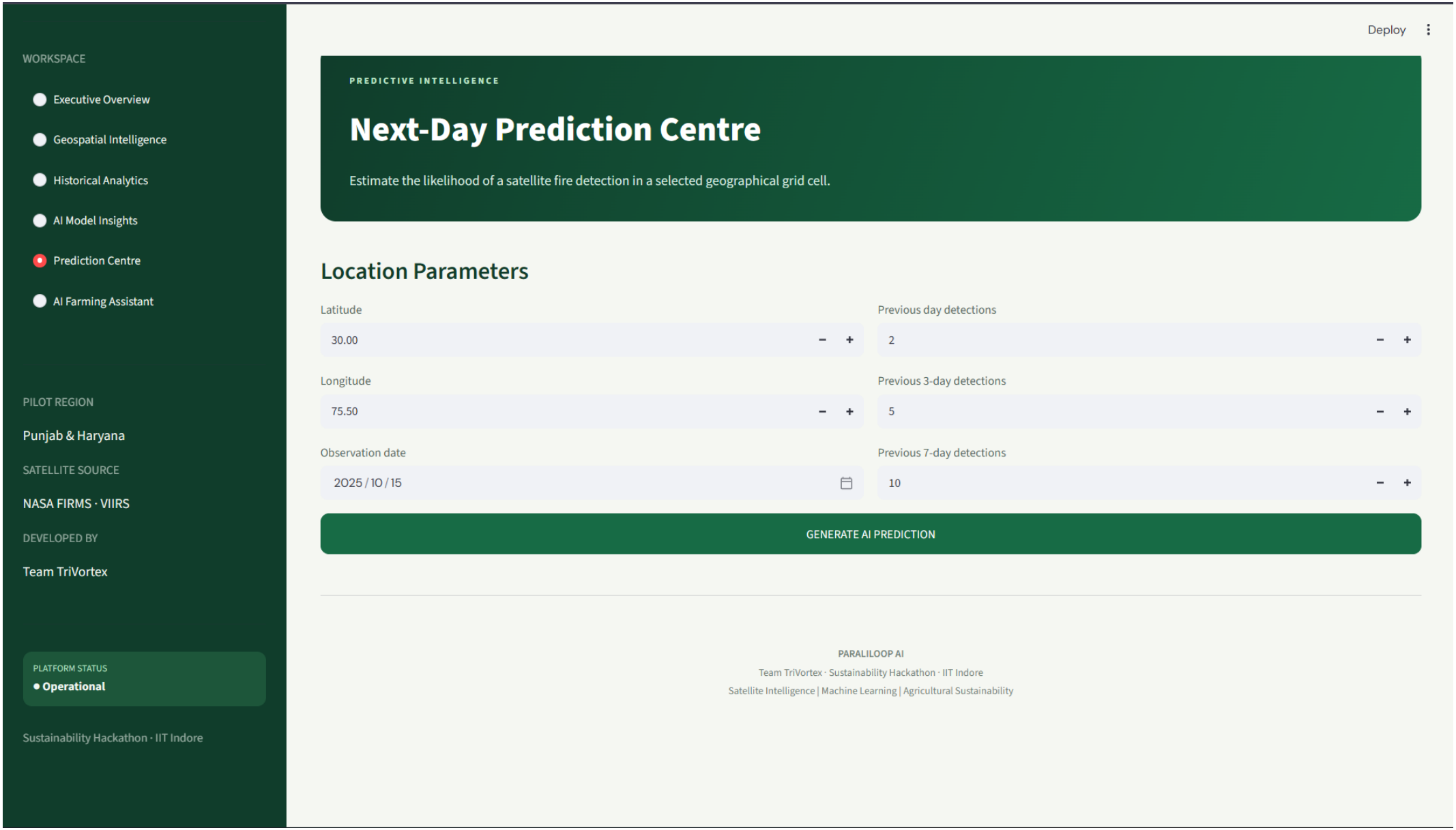
F1 SCORE	ROC-AUC
40.53%	**79.27%**
Precision-recall balance	Class discrimination

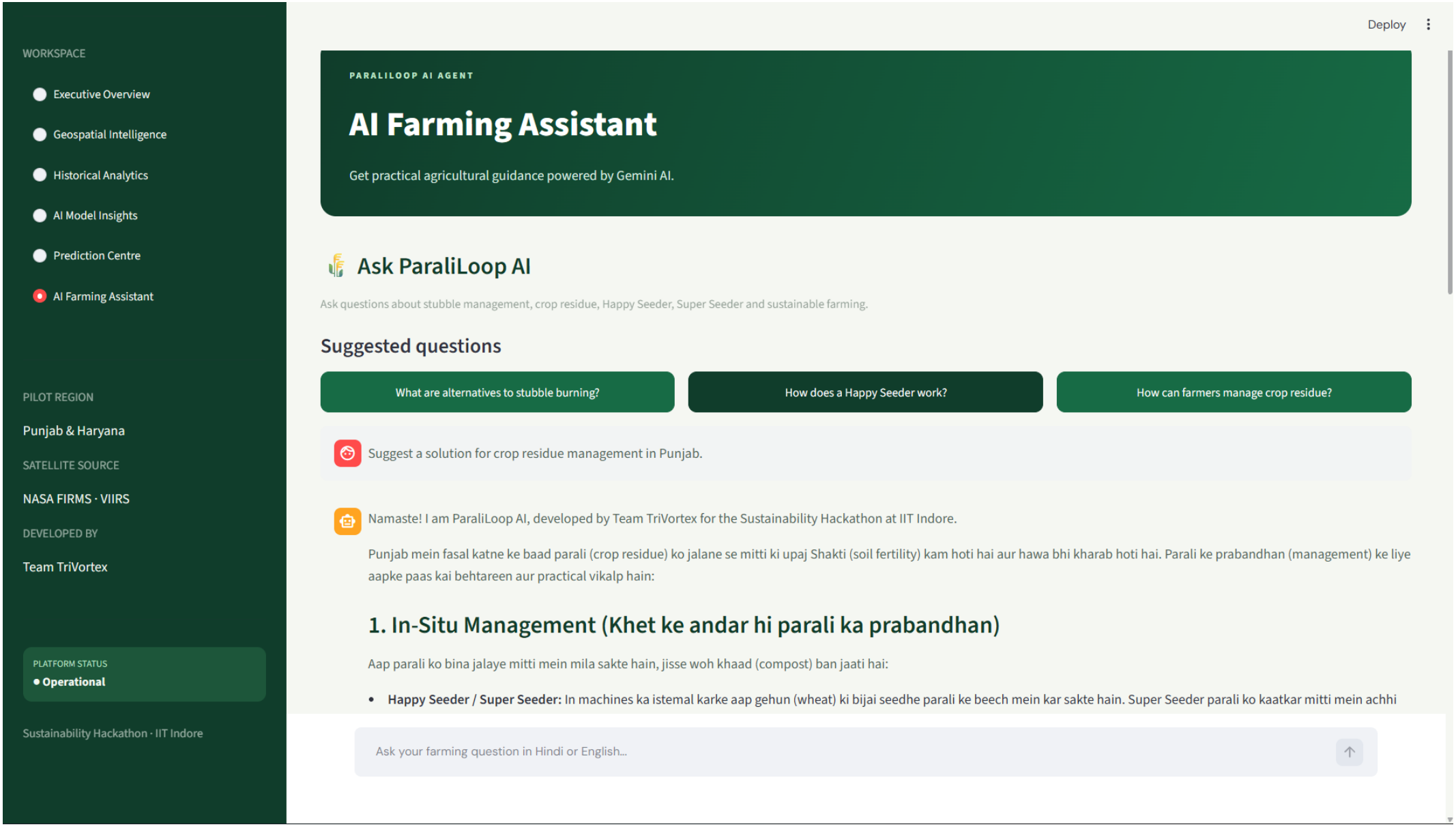
Evaluation Period

Training years: 2021-2024

Testing year: 2025

The current model predicts next-day satellite fire detection. It is not yet a validated crop-residue-burning predictor.


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PREDICTIVE INTELLIGENCE
Next-Day Prediction Centre
Estimate the likelihood of a satellite fire detection in a selected geographical grid cell.
Location Parameters
Latitude
30.00
Longitude
75.50
Observation date
2025 / 10 / 15
Previous day detections
2
Previous 3-day detections
5
Previous 7-day detections
10
GENERATE AI PREDICTION
PARALILOOP AI
Team TriVortex · Sustainability Hackathon · IIT Indore
Satellite Intelligence | Machine Learning | Agricultural Sustainability


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PARALILOOP AI AGENT
AI Farming Assistant
Get practical agricultural guidance powered by Gemini AI.
Ask ParaliLoop AI
Ask questions about stubble management, crop residue, Happy Seeder, Super Seeder and sustainable farming.
Suggested questions
What are alternatives to stubble burning?
How does a Happy Seeder work?
How can farmers manage crop residue?
Suggest a solution for crop residue management in Punjab.
Namaste! I am ParaliLoop AI, developed by Team TriVortex for the Sustainability Hackathon at IIT Indore.
Punjab mein fasal katne ke baad parali (crop residue) ko jalane se mitti ki upaj Shakti (soil fertility) kam hoti hai aur hawa bhi kharab hoti hai. Parali ke prabandhan (management) ke liye aapke paas kai behtareen aur practical vikalp hain:
1. In-Situ Management (Khet ke andar hi parali ka prabandhan)
Aap parali ko bina jalaye mitti mein mila sakte hain, jisse woh khaad (compost) ban jaati hai:
Happy Seeder / Super Seeder: In machines ka istemal karke aap gehun (wheat) ki bijai seedhe parali ke beech mein kar sakte hain. Super Seeder parali ko kaatkar mitti mein achhi
Ask your farming question in Hindi or English...